Venkata Lakshmi Parimala Pasupuleti

AI Enablement Engineer · MCP Servers · Autonomous Agents · Claude Code · LangGraph · Python · TypeScript · AWS · FedRAMP-Adjacent Compliance

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PROFESSIONAL SUMMARY

AI Enablement engineer who builds the tools that make teams AI-native. **MCP server deployments**: multi-provider tool-calling infrastructure routing LLM requests to backend integrations with structured outputs and error handling. **AgentProbe**: open-source automated testing agents and CI regression harness - the exact domain Workiva's AI Enablement team is building. **Leasel**: 8-agent LangGraph autonomous workflows with cloud deployment, durable state, and FedRAMP-adjacent compliance design (OWASP LLM Top 10, HITL gates, audit logging). **Daily Claude Code + Cursor user.** PHI/HIPAA compliance background (100% PHI compliance on AWS S3 migration, Libsodium-encrypted pipelines). Entrata 8yr production scale: 100K+ MAU, microservices, event-driven architecture, CI/CD, mentored 5+ engineers + 600+ students. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Enablement & Agentic AI: MCP (Model Context Protocol), Autonomous Agents, Cloud-Hosted Agents, Agentic AI, AI-Augmented Development, Claude Code, Cursor, LangGraph, LangChain, Multi-Agent Orchestration, LLM Orchestration, Tool Calling, Function Calling, Structured Outputs, RAG Pipelines, Hybrid Search, Semantic Search, Vector Databases, ChromaDB, FastAPI, OpenAI API, Anthropic API, Groq, Ollama, HITL (Human-in-the-Loop)

Testing Agents & LLMOps: Automated Testing Agents, CI/CD Regression Harness, Golden Datasets, LLM Evaluation, LLM-as-Judge, 5-Metric Eval Engine, Hallucination Detection, Observability, Agent Tracing, Agent Sessions, Trace-Level Observability, Metrics Dashboards, Benchmarking, Offline & Online Evaluation, Test Automation, Test-Driven Development (TDD), PDLC

Compliance & Security: FedRAMP-adjacent, PHI/HIPAA Compliance, OWASP LLM Top 10, Audit Logging, Input/Output Filtering, Guardrails, Prompt Injection Defense, SOC 2, NIST, Libsodium, Application Security, Deterministic Validation Gates

Cloud, Backend & DevOps: AWS (S3, EC2, RDS, SQS, Lambda, API Gateway, Bedrock, SageMaker), Docker, Kubernetes, CI/CD (GitHub Actions), Terraform, Microservices, Proxy Servers, Back-End Services, API Integrations, Event-Driven Architecture, RabbitMQ, Redis, Prometheus, Grafana, OpenTelemetry, Observability

Languages & Frameworks: Python (expert), TypeScript, JavaScript, Java, Spring Boot, SQL (expert), Bash, Go (learning); Django, React.js, Node.js, GraphQL; PostgreSQL, MySQL, Oracle, SQLite, Snowflake; Git, Agile/Scrum, Jira

ML & Data: PyTorch, scikit-learn, XGBoost, Hugging Face, NLP, BERT, Anomaly Detection, Feature Engineering, MLflow, Tableau, Power BI

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) — Agentic AI · ML Pipelines · Back-End Integration · AI-Augmented Development

- Developing **agentic AI and ML pipelines** for software applications - integrating AI components with back-end services and APIs; building data pipelines that surface insights into application-facing reporting layers. Applying AI-augmented development practices (Claude Code, Cursor) across the full stack.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — Autonomous Agents · MCP Patterns · LangGraph · Compliance Guardrails · CI/CD

- Built an **autonomous agentic system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - **tool-calling MCP-pattern architecture** routing LLM requests to regulatory corpus and structured data backends; **compliance guardrails** blocking non-compliant outputs before export (FAA altitude, battery, no-fly zone enforcement).
- **RAG layer** over regulatory corpus with cited compliance checks; **automated testing** for guardrail rule verification; **CI/CD pipeline** validating agent behavior on each deploy; **proxy server patterns** for backend integration dispatch.

George Mason University
Aug 2024 – Aug 2025

Research Associate & Teaching Assistant — AI Tooling Evangelism · ML Pipelines · Analytics Engineering

- **Taught Python, AI tooling, and data engineering** to 600+ students across multiple sections (MIS 303) - Python programming, OOP, automation scripting, database design. Built prediction models (Python, R, 190+ countries, $R^2 = 0.977$); fine-tuned classification model (HuggingFace, PyTorch), deployed inference API on Quart. Designed observability dashboards improving research comprehension 40%.

Entrata — Enterprise PropTech SaaS (AWS, 100K+ MAU)
Sep 2018 – Aug 2024

Senior Software Engineer — Back-End Services · API Integrations · AWS Cloud · CI/CD · PHI/HIPAA Compliance

- Owned **back-end services and API integrations** powering 100K+ MAU on AWS - **microservices** (Docker: 50% scalability gain, 30% deployment reduction), **proxy server patterns** and **event-driven async** via RabbitMQ with idempotency guarantees; **CI/CD** (GitHub Actions); AWS S3/EC2/RDS/SQS across 120+ Agile sprints. **Observability**: Prometheus, Grafana, OpenTelemetry; structured logging + incident tracking.
- Achieved **100% PHI/HIPAA compliance** on AWS S3 migration (**FedRAMP-adjacent** compliance posture: Libsodium-encrypted pipelines, OWASP-aligned security design, zero security incidents); **audit logging** across all regulated data workflows; deterministic validation gates blocking silent failures before production; 200+ SQL query optimizations (30% performance gain).
- **AI-augmented development evangelism**: mentored 5+ engineers in tooling adoption and best practices; 300+ peer code reviews; built internal debugging tools reducing debugging time 40% and keeping bug count below 10 for 6+ consecutive months.

Entrata India
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · API Integrations · AWS S3 · PDLC

- Built full-stack enterprise modules (PHP, PostgreSQL, JavaScript, Java); designed SQL import pipelines for 1.6M HUD/Tax Credit records; **AWS S3 API integration** for document storage migration; compliance certification workflows across 56 state-specific configurations; full PDLC ownership from requirements through production.

PROJECTS

AgentProbe: Automated Testing Agents & AI Evaluation Toolkit TESTING AGENTS LLMOPS

Python · LangGraph · FastAPI · Docker · GitHub Actions CI/CD · React · ChromaDB · OpenAI · Anthropic · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Open-source **automated testing agent toolkit** - exactly what Workiva's AI Enablement team is building: **CI/CD-integrated regression harness** with golden datasets blocking deploys on model regressions; **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate); **observability dashboard** (React) tracking per-run metrics, retrieval drift, and latency across sessions. Used to evaluate LeaseI across 1,240+ compliance test cases.
- **Multi-provider LLM integration** (OpenAI, Anthropic, Groq, Ollama) via **MCP-pattern tool-calling runtime** - 3-agent LangGraph orchestration with structured tool dispatch and inter-agent state handoffs; **15-endpoint FastAPI back-end**; Dockerized for consistent deployment; **proxy server pattern** routing LLM requests to appropriate evaluation backends; LLM-as-judge calibration and offline + online evaluation loops.

Leasel: Cloud-Hosted Autonomous Agents — 8-Agent LangGraph, MCP Patterns, HITL, FedRAMP-Adjacent AUTONOMOUS AGENTS MCP

Python · LangGraph · FastAPI · AWS · ChromaDB · OCR · CI/CD · LLM-as-Judge — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **Cloud-hosted autonomous agent platform** on FastAPI + AWS: **8-agent LangGraph pipeline** with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) - **MCP-pattern tool-calling** routing each agent to its backend integration (knowledge stores, OCR layer, structured data retrieval, incident history). **FedRAMP-adjacent compliance design**: OWASP LLM Top 10, audit logging, deterministic validation gates, HITL escalation; every answer cited to exact source clause or flagged unsupported. **93.6% precision, 87.9% F1** on 1,240 lease PDFs (5 US states).
- **Observability**: trace-level agent tracing across all 8 LangGraph nodes; **CI/CD regression gate** blocking production deploys on metric regression; **automated testing harness** (pass@k, LLM-as-judge calibration) used in AgentProbe for cross-validation.

Magizhs Agentic Mission Planner — MCP Tool-Calling, Autonomous Routing, Compliance Guardrails AGENTIC WORKFLOWS

Python · LangGraph · RAG · FastAPI · CI/CD · Test Automation · Proxy Server Patterns

- Stateful LangGraph **autonomous agent system**: natural-language → structured flight plans via **MCP-pattern tool dispatch** (parse → plan → retrieve → validate → conditional routing); **RAG over regulatory corpus** (FAA Part 107) with cited compliance checks; **compliance guardrails** enforcing deterministic safety rules before MAVLink export; automated test suite for guardrail verification in CI/CD.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Agentic AI systems, MCP infrastructure, automated testing agents, LLM evaluation, cloud-native architecture, compliance engineering, data governance

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017